Semantic Segmentation of Liver and Its Tumor Using 3D U-Net

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***Abstract*—In this study, I explore an experimental modification of the U-Net model tailored for segmenting liver and its lesions in contrast-enhanced abdominal CT scans. This adaptation employs a unique combination of Dice Loss and Focal Loss aimed at addressing the challenge of class imbalance between foreground lesions and background voxels. My preprocessing approach involves excluding training examples where the segmentation mask is predominantly background by over 95 percent, thus concentrating the model's learning on clinically significant cases. Validated on the LiTS 2017 dataset, this customized adaptation of the U-Net model illustrates its potential in medical image segmentation. However, it achieved moderate accuracy of 65.34% and Intersection over Union (IoU) of 34.85%, indicating areas for further improvement and optimization in future work.**

***Index Terms*—liver lesion segmentation, U-Net, Dice Loss, Focal Loss, class imbalance, high-resolution medical images, LiTS 2017 challenge.**

I. Introduction

Liver tumor segmentation from medical imaging is a critical step in the diagnosis, treatment planning, and monitoring of liver diseases. Precise segmentation enables clinicians to assess the size, shape, and progression of tumors, which is essential for making informed decisions regarding patient care. However, this task is fraught with challenges due to the heterogeneous appearance of liver tumors, their irregular shapes, and the similarity in appearance between tumors and surrounding tissues. These factors contribute to the difficulty of developing automated segmentation methods that can accurately differentiate between healthy and diseased tissue.

The Liver Tumor Segmentation Challenge (LiTS) 2017 (LiTS 2017 challenge) presented a significant opportunity for researchers to tackle these issues head-on. It provided a standardized, high-quality dataset of contrast-enhanced abdominal CT scans, aiming to benchmark the performance of various segmentation algorithms and to foster advancements in medical imaging analysis. Over the years, numerous methods have been applied to this dataset, ranging from traditional image processing techniques to advanced deep learning models. Among these, convolutional neural networks (CNNs) have emerged as a particularly promising avenue due to their ability to learn complex patterns in imaging data.

The U-Net architecture, initially designed for biomedical image segmentation [1], has received widespread acclaim for its effectiveness in this domain. Its unique structure, characterized by a symmetric expanding path that enables precise localization, has been pivotal in advancing the field. The introduction of 3D U-Net extended this architecture to three-dimensional image data, offering an enhanced ability to capture the volumetric context of medical scans, which is crucial for accurately segmenting tumors in organs such as the liver.

This project employs the 3D U-Net model to segment liver tumors from the LiTS 2017 dataset. Rather than introducing new methodologies, my work focuses on demonstrating the effectiveness of 3D U-Net for multi-class semantic segmentation in biomedical imaging. By adapting this architecture to the specific challenges presented by liver tumor segmentation, I aim to showcase its capabilities in handling the intricacies of high-resolution, three-dimensional medical scans. Through my application of 3D U-Net, I contribute to the body of evidence supporting its use as a robust tool for medical image analysis, particularly in the critical context of liver disease diagnosis and treatment planning.

II. Literature Review

In their innovative study, "Training on Polar Image Transformations," [2]the authors introduce a novel neural network method using polar transformations to significantly improve medical image segmentation, especially for elliptically shaped structures like organs. Their technique simplifies segmentation by reducing dimensionality and clearly separating segmentation from localization, proposing two methods to determine the optimal polar origin. Their approach demonstrated state-of-the-art performance, winning the LiTS 2017 challenge with an Intersection over Union (IoU) of 89.85% and a Dice coefficient of 93.02%. In their paper [3] on KiU-Net, the second-place winner of the LiTS 2017 challenge, the authors address the shortcomings of traditional encoder-decoder models in medical image segmentation by proposing a novel architecture that combines an overcomplete convolutional approach with traditional U-Net to enhance the detection of small structures and precise segmentation of boundary regions. This innovative method not only showed significant improvement across various datasets but also achieved a notable Intersection over Union (IoU) score of 89.46. In another noteworthy approach

[4] The authors introduce Semantic Genesis, a self-supervised 3D model that enhances medical imaging analysis by leveraging inherent anatomical patterns, outperforming traditional methods and securing third with an IoU of 85.6 and a Dice coefficient of 92.27.Further advancing the exploration of deep learning in medical imaging, the next study [5] Models Genesis, developed for 3D medical image analysis through self-supervision, outperform traditional methods by maintaining crucial three-dimensional data, securing fourth in the LiTS 2017 challenge with an IoU of 79.52 and a Dice coefficient of 91.13.The PVTFormer [6] combines Vision Transformers and advanced techniques, efficiently addressing scalability and computational challenges in high-resolution medical image analysis. It achieved a Dice coefficient of 86.78, an IoU of 78.46, and a Hausdorff Distance of 3.50 in the LiTS 2017 challenge, ranking fifth.

Continuing the exploration of cutting-edge approaches in medical imaging, particularly in liver segmentation from CT scans, the latest study H-DenseUNet, a novel hybrid model, [7] efficiently merges 2-D and 3-D CNN strengths for enhanced feature extraction and volumetric context aggregation in medical image analysis. It secured the sixth position in the LiTS 2017 challenge with a notable Dice coefficient of 96.5.another innovative approach [8] At the LiTS 2017 competitions is a novel segmentation method featuring cascaded, densely connected 2D U-Nets and a Tversky-coefficient loss function was introduced, improving liver lesion detection from CT scans. This fully-automatic pipeline offers adjustable hyperparameters for enhanced adaptability and sensitivity. It secured the eightieth position in the LiTS 2017 challenge with a Dice coefficient of 94.The MSFF-Net, [9], leveraging 3D V-Net architecture for liver segmentation, overcomes challenges such as low contrast and variable shapes through innovative modules, showing marked improvements with Dice scores of 0.963 and 0.964 on the LiTS 2017 dataset.The study [10] presents an Attention Hybrid Connection Network for liver tumor segmentation, integrating attention mechanisms with skip connections, showing effectiveness with a 59.1% dice score on LiTS, 3DIRCADb, and Clinical datasets.The enhanced U-Net architecture, mU-Net, [11] introduces a residual path and extra convolution layers for improved liver and tumor segmentation, addressing traditional U-Net limitations. On the LiTS 2017 dataset, mU-Net achieved a liver-tumor Dice similarity coefficient of 89.72% and a liver segmentation DSC of 98.51%. For the 3Dircadb dataset, Dice coefficients were 96.01% for liver and 68.14% for liver-tumor segmentations, showcasing its advanced performance.

The review of recent literature on liver tumor segmentation demonstrates significant advancements in leveraging deep learning techniques to enhance the precision and efficiency of medical imaging analysis. Innovations such as the incorporation of multi-scale feature fusion, attention mechanisms, and novel U-Net architectures have shown promising results in overcoming the inherent challenges of segmenting liver and liver tumors, characterized by low tissue contrast and variable shapes.

III. LiTS 2017- Liver Tumor Segmentation Challenge

The liver is a common site of primary (i.e. originating in the liver like hepatocellular carcinoma, HCC) or secondary (i.e. spreading to the liver like colorectal cancer) tumor development. Due to their heterogeneous and diffusive shape, automatic segmentation of tumor lesions is very challenging. The data and segmentations are provided by various clinical sites around the world. The training data set contains 130 CT scans and the test data set 70 CT scans.(LiTS 2017 challenge).

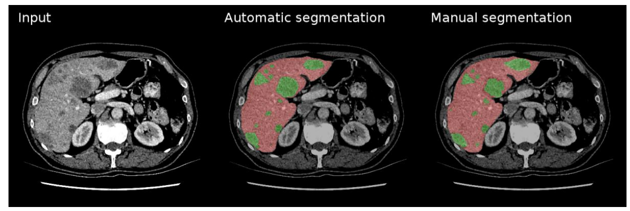

Fig. 1. Example of segmentation output compared to ground truth ("Manual segmentation"). Lesions in green, liver in red. [12]

IV. Problem Statement

Given the complexities and objectives outlined for the Liver Tumor Segmentation (LiTS) dataset challenge, which encompasses the accurate segmentation of the liver and liver tumors from CT images, the development of robust and clinically applicable algorithms remains a formidable task. This project addresses the inherent difficulties of liver and tumor segmentation, characterized by the necessity to differentiate the liver from surrounding anatomical structures and the challenge of accurately detecting and delineating liver tumors, which exhibit significant variability in size, shape, and contrast. Furthermore, the project aims to contribute to the body of knowledge on the effectiveness of the 3D U-Net architecture for multi-class semantic segmentation within this context. By focusing on adapting the 3D U-Net to meet the specific segmentation challenges presented by the LiTS dataset, including ensuring robustness and generalizability across diverse image sets, this study seeks to validate the potential of 3D U-Net as a powerful tool for medical image analysis in liver disease diagnosis and treatment planning.

A. Challenges

In working with the LiTS dataset for liver and tumor segmentation, several intrinsic challenges emerged, reflecting the complex nature of medical image analysis. The dataset's inherent variability in liver and tumor appearance posed a significant hurdle, requiring a model capable of generalizing across diverse manifestations of liver shapes, sizes, and tumor presentations. This task was further complicated by the often low contrast between liver tissues and tumors, where distinguishing between the two becomes a nuanced challenge, particularly in the presence of noise and artifacts common in CT imaging. Moreover, the proximity of the liver to other organs with similar radiodensity added an extra layer of complexity in accurately delineating the liver's boundaries. A particularly vexing issue was the significant class imbalance between foreground lesions and background voxels. The predominance of background voxels can lead models to become biased towards identifying non-lesion areas, potentially overlooking smaller or less distinct lesions. To address this, a preprocessing strategy was implemented to exclude training examples where the segmentation mask was predominantly background by over 95 percent. This approach effectively focused the model's learning on clinically significant cases, mitigating the impact of class imbalance and ensuring a more balanced learning environment. This preprocessing step was pivotal in enhancing the model's ability to accurately segment liver lesions, showcasing the necessity of tailored approaches in overcoming dataset-specific challenges.

A significant challenge encountered was the substantial computational resource requirements. The large dimensions of CT scans in the dataset necessitated an extensive amount of system memory, often pushing beyond the limits of what standard computational resources could provide. This issue was particularly pronounced when working with full-resolution data, where the memory demands could escalate to require between 12 to 51 GB of system RAM. The challenge of balancing the need for high-resolution data to capture essential anatomical details against the constraints of available computational hardware was a constant hurdle. Without employing specific mitigation strategies, the task of processing and analyzing the dataset posed a considerable bottleneck, highlighting the critical impact of computational resource limitations on medical image analysis projects.

V. Methods

A. Preprocessing

The dataset was downloaded and stored in Google Drive, which was later mounted on Google Colab Pro, as part of the preprocessing steps. In addressing the challenges posed by the LiTS dataset, a two-pronged preprocessing strategy was meticulously implemented. Firstly, to tackle the pervasive issue of class imbalance between foreground lesions and background voxels, a selective exclusion approach was adopted. This involved filtering out training examples where the segmentation mask was overwhelmingly composed of background (over 95 percent), thereby honing the model's focus on clinically significant cases. This effort was further augmented by the innovative application of a combined Dice Loss and Focal Loss optimization, specifically engineered to address the imbalance challenge by sensitizing the model to both lesion presence and the nuanced differentiation of lesions from the background. Secondly, the considerable challenge of computational resource requirements was approached through strategic data reshaping. By resizing the dataset images to 256x256x64, taking slices from the central regions of each scan (specified as 128:384, 128:384, 1:65), the model's operational memory demands were significantly reduced. This resizing enabled the handling of data within a manageable 12 to 51 GB of system RAM range, demonstrating a careful balance between computational efficiency and the preservation of critical image details necessary for accurate segmentation. This dual-faceted preprocessing approach effectively mitigated the key challenges inherent in the LiTS dataset, paving the way for improved model performance and efficiency.

B. Model

The model employed is a 3D U-Net architecture, tailored for volumetric data analysis, such as CT scans. This 3D extension of the U-Net introduced by Ronneberger et al. in 2015 [1] is constructed with a series of 3D convolutional layers (Conv3D), demonstrating the model's capability to process and analyze three-dimensional data. It follows a symmetrical design, featuring a contraction path to capture context and an expansive path for precise localization. Each convolutional layer in the contraction path is followed by a dropout layer, with dropout rates ranging from 0.1 to 0.3, to prevent overfitting and ensure generalization. The layers utilize 'ReLu' activation functions, promoting non-linear transformations within the network, while the final output layer applies a 'Softmax' activation function, ideal for the multi-class segmentation by providing a probabilistic distribution over different classes. The model also integrates MaxPooling3D for downscaling and Conv3DTranspose layers for upscaling, facilitating effective feature extraction and resolution restoration. The network architecture is depicted in Figure 2. To address the challenge of class imbalance, a novel combination of Dice Loss and Focal Loss is optimized, focusing on learning from clinically significant cases. This model embodies a careful balance of complexity and performance, tailored to the specific challenges of the LiTS dataset.

The model is programmed using the Keras API with the Tensorflow backend in Python. The model hyper parameters are given in Table I. For the computational work required in this research, the model was trained and executed on Colab Pro, a cloud-based service that offers advanced computational resources. Utilizing Colab Pro allowed for the employment of a T4 GPU alongside 51GB of system RAM, which significantly enhanced the capability to train the deep learning model efficiently. The robust computing environment provided by Colab Pro was crucial for meeting the demanding requirements of training a 3D U-Net model, offering substantial improvements in processing power and execution speed compared to more conventional setups.

C. Loss

In the development of the segmentation model, a critical aspect of optimization involved addressing the challenge of class imbalance. This was achieved through the application of a combined Dice Loss and Focal Loss strategy.

Dice Loss, often utilized in medical image segmentation, measures the overlap between the predicted segmentation and the ground truth. The formula for Dice Loss is given by:

$$\text{Dice Loss} = 1 - \frac{2 \times |Y \cap \hat{Y}|}{|Y| + |\hat{Y}|}$$

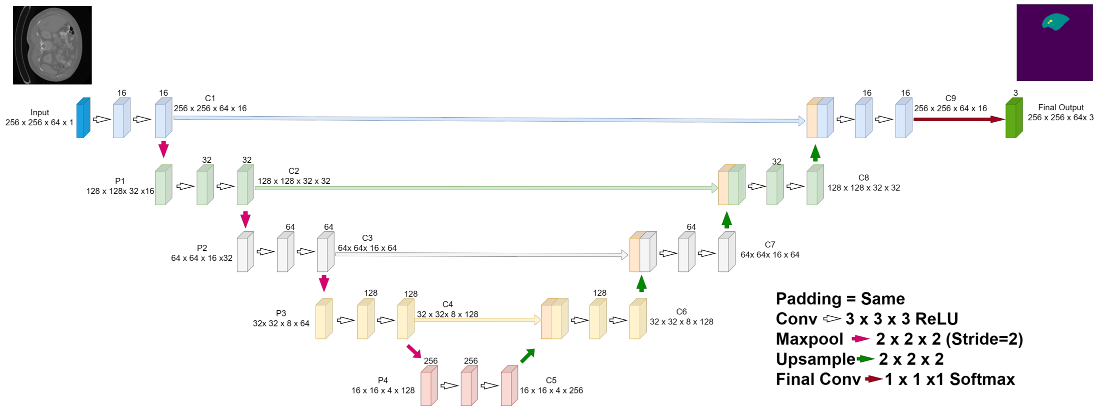

Fig. 2. 3D U-Net For Liver Tumor Segmentation

where Y represents the ground truth mask, and $\hat{Y}$ represents the predicted segmentation mask.

Focal Loss, designed to address class imbalance by modifying the standard cross-entropy loss such that it down-weights the loss assigned to well-classified examples. Its formula is expressed as:

$$\text{Focal Loss} = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

where p_t is the model's estimated probability for each class, γ is the focusing parameter that adjusts the rate at which easy examples are down-weighted, and α_t is a weighting factor for each class to counteract class imbalance further.

The combined Dice and Focal Loss strategy leverages the strengths of both losses: Dice Loss for its effectiveness in segmentation overlap and Focal Loss for its capacity to manage class imbalance. The combined loss function is formulated as the sum of Dice Loss and Focal Loss, with equal weighting applied to both, It underscores a balanced approach towards enhancing segmentation accuracy while simultaneously correcting for class imbalance. This innovative combination ensures the model is finely attuned to both the presence of lesions and their differentiation from the background, significantly boosting the model's efficacy on the LiTS dataset.

D. Evaluation Metrics

To comprehensively assess the performance of the segmentation model on the LiTS dataset, two pivotal metrics were employed: Accuracy and Mean Intersection over Union (IoU).

Accuracy serves as a primary metric to gauge the overall effectiveness of the model in classifying pixels correctly across the entire dataset. It is defined as the ratio of correctly predicted pixels to the total number of pixels, encompassing both lesion and non-lesion regions.

Mean Intersection over Union (Mean IoU), on the other hand, offers a more nuanced evaluation, especially valuable in segmentation tasks. IoU, also known as the Jaccard Index, measures the overlap between the predicted segmentation and the ground truth. For a single class, IoU is calculated as:

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|}$$

where A represents the predicted segmentation area, and B denotes the ground truth area. The Mean IoU is then derived by averaging the IoU scores across all classes, providing a comprehensive metric that reflects both the precision and recall of the model's segmentation.

VI. Results

In this project, the primary objective was to showcase the efficacy of the 3D U-Net architecture for multi-class semantic segmentation within the context of biomedical imaging, specifically for segmenting liver tumors from the LiTS 2017 dataset. The study did not aim to introduce novel methodologies but rather to demonstrate the application and performance of an established model in accurately delineating complex anatomical structures. Performance metrics reveal a mixed outcome: while The model attained an accuracy of 0.65, signifying a moderate capability in correctly classifying pixels or voxels as either tumor or non-tumor, the Intersection over Union (IoU) score was relatively low at 0.34. This suggests that, despite correctly classifying a high percentage of the area, the precision of segmentation—particularly the overlap between predicted and actual tumor regions—was limited. These results underscore the challenges in balancing overall accuracy with precise localization in volumetric segmentation tasks and suggest avenues for further refinement of the model and its training process. Table II summarizing the key performance metrics.

Hyperparameter	Value
Epochs	20
Batch Size	1
Optimizer	Adam
Learning Rate	0.0001
Trainable Parameters	5,644,947

TABLE I
HYPERPARAMETERS OF THE 3D U-NET MODEL

In evaluating the model's performance, I closely monitored the progression of training and validation accuracy, as well as loss across epochs. Figure 3 depict the model's training and validation accuracy over epochs, demonstrating a consistent

Metric	Value
Accuracy	0.65
IoU Score	0.34

TABLE II
PERFORMANCE METRICS OF THE 3D U-NET MODEL ON THE LITS 2017 DATASET

improvement in accuracy as training progresses. Figure 4, on the other hand, illustrate the training and validation loss over epochs, showcasing a decrease in loss values, indicative of the model's effective learning and optimization over time. These plots are instrumental in visualizing the model's ability to generalize and optimize throughout the training process.

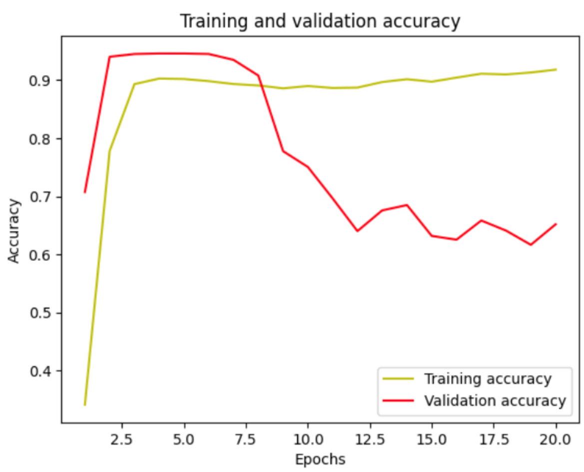

Fig. 3. Training and Validation Accuracy vs Epoch.

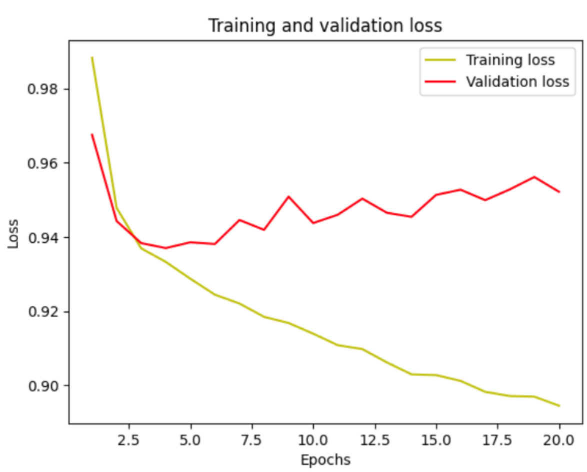

Fig. 4. Training and Validation Loss vs Epoch

VII. CONCLUSION

In this project, while the model did not achieve scores comparable to the winners of the LiTS 2017 challenge, it nonetheless demonstrated the potential and effectiveness of the 3D U-Net architecture for multi-class semantic segmentation in biomedical imaging. By tailoring this sophisticated model to address the unique demands of liver tumor segmentation, the study aims to highlight the 3D U-Net's adeptness in managing the complexities of high-resolution, three-dimensional medical scans. The endeavor contributes valuable insights to the ongoing discourse on leveraging 3D U-Net as a formidable tool for medical image analysis, especially pertinent in diagnosing and planning treatment for liver diseases. However, the journey unveiled the inherent challenges of preprocessing, training duration, and constraints imposed by physical memory limitations, which somewhat hindered achieving optimal results. These findings signal critical areas for potential enhancements in model training and optimization strategies, underscoring the delicate task of balancing accuracy with precise localization within volumetric segmentation endeavors. This experience not only charts pathways for refining the model and its training regimen but also fortifies the case for 3D U-Net's applicability in the nuanced sphere of medical imaging, with the prospect of surmounting current limitations through focused improvements and technological advancements.

REFERENCES

[1] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in Computer Science Department and BIOSS Centre for Biological Signalling Studies, University of Freiburg, Germany, 2015.
[2] Marin Bencevic, Irena Galic, Marija Habijan, and Danilo DAbin,"Training on Polar Image Transformations Improves Biomedical Image Segmentation," Faculty of Electrical Engineering, Computer Science and Information Technology, J.J. Strossmayer University of Osijek, Croatia, 2021.
[3] Jeya Maria Jose Valanarasu, Vishwanath A. Sindagi, llker Hacihaliloglu, Vishal M. Patel, "KiU-Net: Overcomplete Convolutional Architectures for Biomedical Image and Volumetric Segmentation," 2021
[4] Fatemeh Haghighi, Mohammad Reza Hosseinzadeh Taher, Zongwei Zhoi, Michael B. Gotway, Jianming Liang,"Learning Semantics-enriched Representation via Self-discovery, Self-classification, and Self-restoration," Arizona State University, USA 2020
[5] Zongwei Zhou, Vatsal Sodha, Md Mahfuzur Rahman Siddiquee, Ruibin Feng, Nima Tajbakhsh, Michael B. Gotway, Jianming Liang,"Models Genesis: Generic Autodidactic Models for 3D Medical Image Analysis", Arizona State University, 2019
[6] Debesh Jha, Nikhil Kumar Tomar, Koushik Biswas, Gorkem Durak, Alpay Medetalibeyoglu, Mathew Antalek, Yury Velichko, Daniela Ladner, Amir Borhani, Ulas Bagci, "CT Liver Segmentation Via PVT-BAsed Encoding And Refined Decoding", Machine and Hybrid Intelligence Lab, Department of Radiology, Northwestern University, 2023
[7] Xiaomeng Li, Hao Chen, Xiaojuan Qi, Qi Dou, Chi-Wing Fu, and Pheng-Ann Heng, "H-DenseUNet:Hybrid Densely connected UNet for Liver and Tumor Segmentation From CT Volumes", 2017
[8] Karsten Roth, Tomasz Konopczynski, Jurgen Hesser, "Liver Lesion Segmentation with Slice-Wise 2D Tiramisu And Tversky Loss Function", Experimental Radiation Oncology, Department of Radiation Oncology, University Medical Center Mannheim, Heidelberg University, Germany, 2019.
[9] Yingxin Zhou, Jiashuai Zong,"Automatic Liver Segmentation Method from CT Images Based on Improved 3D U-Net", College of Computer Science and Engineering, Shandong University of Science and Technology, Qindao, China, 2022
[10] Huiyan Jiang, Tianyu Shi, Zhiqi Bai, and Liangliang Huang,"AHCNet:An application of Attention Mechanism and Hybrid Connection for Liver Tumor Segmentation in CT Volumes", Sofware College, Northeastern University, Shenyang , China, 2019.

[11] Hyunseok Seo, Charles Huang, Maxime Bassenne, Ruoxiu Xiao, and Lei Xing,"Modified U-Net(mU-Net) With Incorporation of Object-Dependent High Level Features for Improved Liver and Liver-Tumor Segmentation in CT Images",2020.
[12] Eugene Vorontsov, An Tang, Chris Pal, and Samuel Kadoury, "Liver Lesion Segmentation Informed By Joint Liver Segmentation", Montreal Institute for Learning Algorithms (MILA),2018.